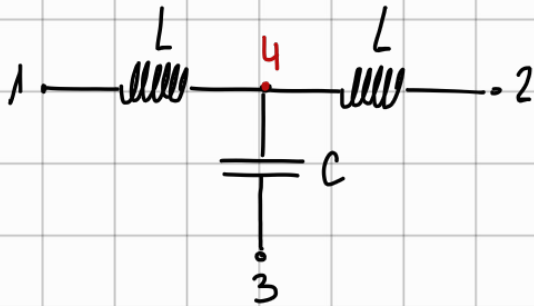
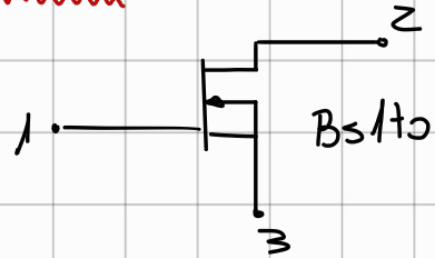


Trabajo semanal 5

Caso 1



Caso 2



1)

Matriz de admitancia indefinida.

Caso 1

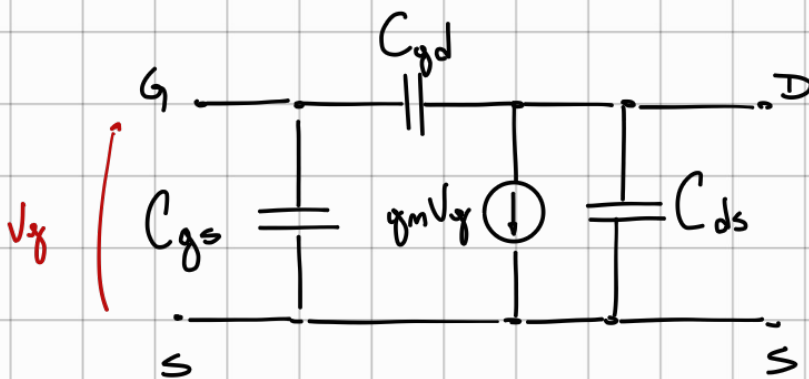
$$Y_{Hai,1} = \begin{pmatrix} Y_L & \emptyset & \emptyset & -Y_L \\ \emptyset & Y_L & \emptyset & -Y_L \\ \emptyset & \emptyset & Y_C & -Y_C \\ -Y_L & -Y_L & -Y_C & 2Y_L + Y_C \end{pmatrix} = \begin{pmatrix} 1/S_L & \emptyset & \emptyset & -1/S_L \\ \emptyset & 1/S_L & 0 & -1/S_L \\ \emptyset & \emptyset & SC & -SC \\ -1/S_L & -1/S_L & -SC & 2/S_L + SC \end{pmatrix}$$

Caso 2

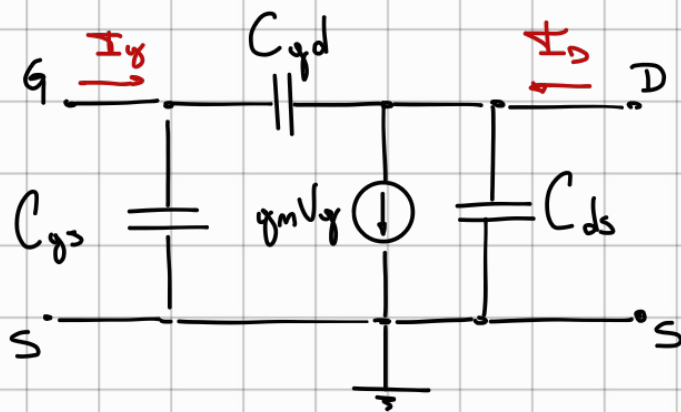
De la hoja de datos:

- * $C_{iss} : C_{gs} + C_{gd}$
- * $C_{oss} : C_{ds} + C_{gd}$
- * $C_{rss} : C_{gd}$

Circuito equivalente



4 Configuración source común.



En términos de sus parámetros de admitancia

$$I_1 = sC_{gs} V_g + sC_{gd} (V_g - V_D)$$

$$I_1 = V_g \underbrace{s(C_{gs} + C_{gd})}_{Y_{11}} - V_D \underbrace{(sC_{gd})}_{Y_{12}}$$

$$I_1 = I_g, I_2 = I_d$$

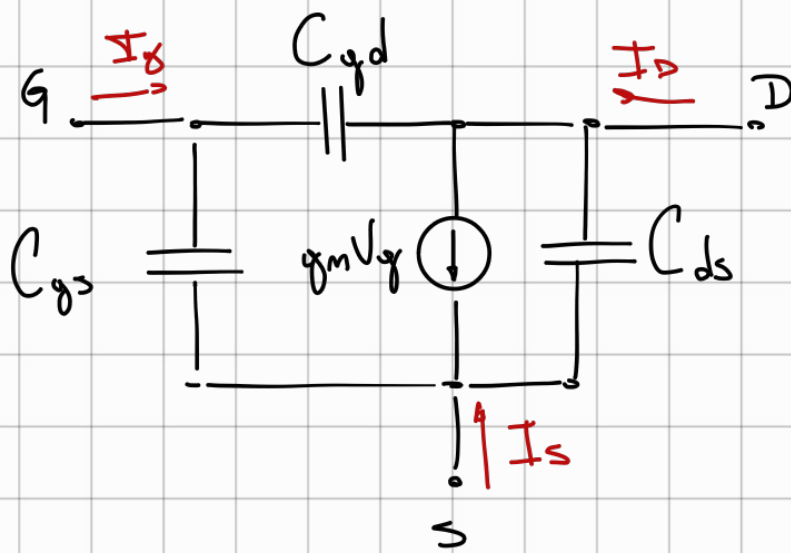
$$I_2 = V_g \cdot g_m + sC_{gd} (V_D - V_g) + V_D sC_{ds}$$

$$I_2 = V_g \underbrace{(g_m - sC_{gd})}_{Y_{21}} + V_D \underbrace{s(C_{gd} + C_{ds})}_{Y_{22}}$$

$$\begin{pmatrix} I_g \\ I_d \end{pmatrix} = \begin{pmatrix} s(C_{gs} + C_{gd}) & -sC_{gd} \\ g_m - sC_{gd} & s(C_{ds} + C_{gd}) \end{pmatrix} \begin{pmatrix} V_g \\ V_D \end{pmatrix}$$

$$\begin{pmatrix} I_g \\ I_d \end{pmatrix} = \begin{pmatrix} sC_{iss} & -sC_{gd} \\ g_m - sC_{gd} & sC_{oss} \end{pmatrix} \begin{pmatrix} V_g \\ V_D \end{pmatrix} \quad \text{Matriz de admitancias}$$

Ahora podemos hallar la MAT, sumando la fila y la columna correspondiente al termino source flotante.



$$\begin{pmatrix} I_g \\ I_d \\ I_s \end{pmatrix} = \begin{pmatrix} sC_{gs} & -sC_{rss} & s(C_{rss} - C_{rss}) \\ g_m - sC_{rss} & sC_{oss} & s(C_{rss} - C_{oss}) - g_m \\ s(C_{rss} - C_{rss}) - g_m & s(C_{rss} - C_{oss}) & s(C_{rss} - C_{rss}) + s(C_{oss} - C_{rss}) + g_m \end{pmatrix} \begin{pmatrix} V_g \\ V_d \\ V_s \end{pmatrix}$$

* Matriz de admitancia Indefinida para el Mosfet.

2)

a) $A_{13}^{23} = \frac{V_{23}}{V_{13}} = \frac{V_{ij}}{V_{mn}} \quad m=1 \quad i=2 \quad Y_{ij}^{mn}$
 $n=3 \quad j=3$

$$Y_{23}^{13} = (-1) \begin{vmatrix} \phi & -1/sL \\ -1/sL & \frac{2}{sL} + sC \end{vmatrix} = \frac{1}{(sL)^2}$$

~~$$\begin{pmatrix} 1/sL & \phi & \phi & -1/sL \\ \phi & 1/sL & \phi & -1/sL \\ \phi & \phi & sC & -sC \\ -1/sL & -1/sL & -sC & 2/sL + sC \end{pmatrix}$$~~

$$Y_{13}^{13} = \begin{vmatrix} \frac{1}{sL} & -\frac{1}{sL} \\ -\frac{1}{sL} & \frac{2}{sL} + sC \end{vmatrix} = \frac{2}{(sL)^2} + \frac{C}{L} - \frac{1}{(sL)^2}$$

$$Y_{13}^{13} = \frac{s^2 LC + 1}{(sL)^2}$$

~~$$\begin{pmatrix} 1/sL & \phi & \phi & -1/sL \\ \phi & 1/sL & \phi & -1/sL \\ \phi & \phi & sC & -sC \\ -1/sL & -1/sL & -sC & 2/sL + sC \end{pmatrix}$$~~

$$\frac{V_{23}}{V_{13}} = \frac{\frac{1}{(sL)^2}}{\frac{s^2 LC + 1}{(sL)^2}} \rightarrow \boxed{\frac{V_{23}}{V_{13}} = \frac{1}{LC} \frac{1}{s^2 + 1/LC}}$$

$$\star A_{31}^{21} = \frac{V_{21}}{V_{31}} = \frac{V_{ij}}{V_{mn}} \quad \begin{matrix} m=3 & i=2 \\ n=1 & j=1 \end{matrix}$$

$$\underline{Y}_{21}^{31} = (-1) \begin{vmatrix} 0 & -1/sL \\ -sC & \frac{2}{sL} + sC \end{vmatrix} = \frac{C}{L}$$

$$\begin{pmatrix} 1/sL & \emptyset & \emptyset & -1/sL \\ \emptyset & 1/sL & \emptyset & -1/sL \\ \emptyset & \emptyset & sC & -sC \\ -1/sL & -1/sL & -sC & 2/sL + sC \end{pmatrix}$$

$$\underline{Y}_{31}^{31} = \begin{vmatrix} 1/sL & -1/sL \\ -1/sL & \frac{2}{sL} + sC \end{vmatrix} = \frac{1}{(sL)^2} \frac{C}{L} - \frac{1}{(sL)^2} = \frac{1}{(sL)^2} + \frac{C}{L}$$

$$\begin{pmatrix} 1/sL & \emptyset & \emptyset & -1/sL \\ \emptyset & 1/sL & \emptyset & -1/sL \\ \emptyset & \emptyset & sC & -sC \\ -1/sL & -1/sL & -sC & 2/sL + sC \end{pmatrix}$$

$$\underline{Y}_{31}^{31} = \frac{1 + s^2 CL}{(sL)^2}$$

$$V_{31}^{21} = \frac{\frac{C}{L}}{\frac{1 + s^2 CL}{(sL)^2}} = \frac{s^2}{s^2 + 1/LC} \rightarrow \boxed{\frac{V_{21}}{V_{31}} = \frac{s^2}{s^2 + 1/LC}}$$

$$\star A_{32}^{12} = \frac{V_{12}}{V_{32}} = \frac{V_{ij}}{V_{mn}} \quad \begin{matrix} m=3 & i=1 \\ n=2 & j=2 \end{matrix}$$

$$\underline{Y}_{12}^{32} = \begin{vmatrix} \emptyset & -1/sL \\ -sC & \frac{2}{sL} + sC \end{vmatrix} = -\frac{C}{L}$$

$$\begin{pmatrix} 1/sL & \emptyset & \emptyset & -1/sL \\ \emptyset & 1/sL & \emptyset & -1/sL \\ \emptyset & \emptyset & sC & -sC \\ -1/sL & -1/sL & -sC & 2/sL + sC \end{pmatrix}$$

$$Y_{32}^{32} = \begin{vmatrix} \frac{1}{sL} & -\frac{1}{sL} \\ -\frac{1}{sL} & \frac{2}{sL} + sC \end{vmatrix} = \frac{2}{(sL)^2} + \frac{C}{L} - \frac{1}{(sL)^2}$$

$$Y_{32}^{32} = \frac{1}{(sL)^2} + \frac{C}{L} = \frac{s^2 CL + 1}{(sL)^2}$$

$$\frac{V_{12}}{V_{32}} = \frac{-C/L}{\frac{s^2 CL + 1}{(sL)^2}} = -\frac{s^2 LC}{s^2 CL + 1}$$

$$\frac{V_{12}}{V_{32}} = \frac{s^2}{s^2 + 1/LC}$$

$$\begin{pmatrix} \frac{1}{sL} & 0 & 0 & -\frac{1}{sL} \\ 0 & \frac{1}{sL} & 0 & -\frac{1}{sL} \\ 0 & 0 & sC & -sC \\ -\frac{1}{sL} & -\frac{1}{sL} & -sC & \frac{2}{sL} + sC \end{pmatrix}$$

b) tenemos:

$$\frac{V_{23}}{V_{13}} = \frac{1}{LC} \frac{1}{s^2 + 1/LC} = t_a$$

$$\frac{V_{21}}{V_{31}} = \frac{s^2}{s^2 + 1/LC} = t_b$$

$$\frac{V_{12}}{V_{32}} = \frac{s^2}{s^2 + 1/LC} = t_c$$

$$t_b = t_c$$

$$|t_a + t_b| = 1$$

$$\left| \frac{1/LC}{s^2 + 1/LC} + \frac{s^2}{s^2 + 1/LC} \right| = 1$$

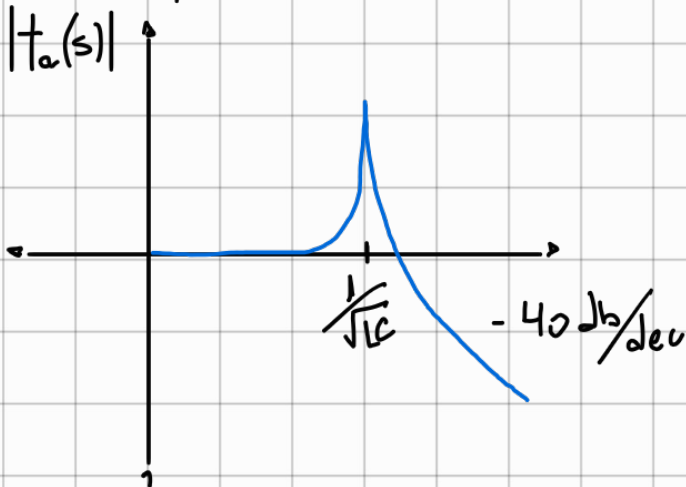
$$\left| \frac{s^2 + 1/LC}{s^2 + 1/LC} \right| = 1$$

t_b y t_c son complementarios en módulo de t_a

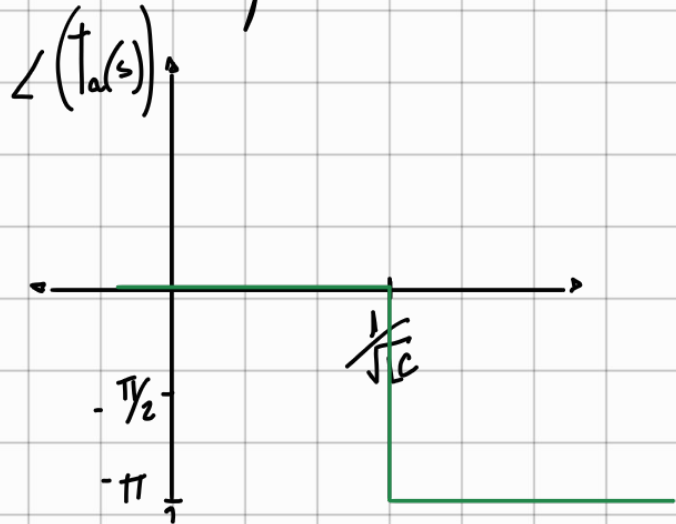
Respuesta de módulo y fase.

$$* T_a = \frac{V_{23}}{V_{13}} = \frac{1}{LC} \frac{1}{s^2 + 1/LC}$$

Respuesta de módulo.

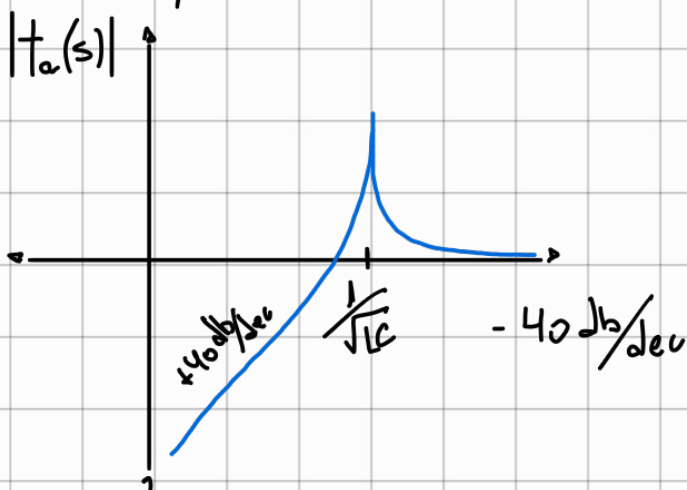


Respuesta de fase.

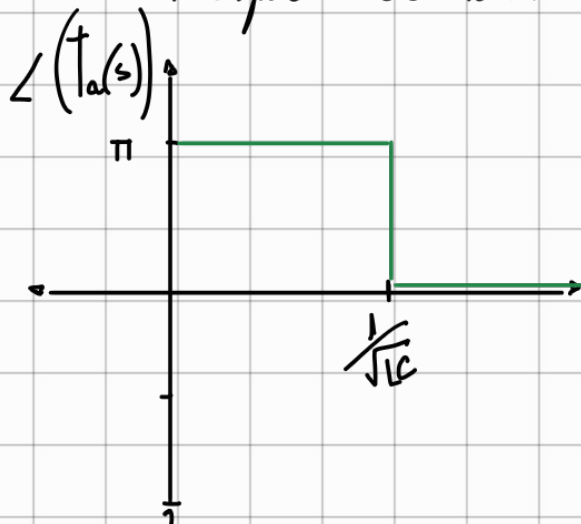


$$* T_b = T_c = \frac{V_{23}}{V_{13}} = \frac{s^2}{s^2 + 1/LC}$$

Respuesta de módulo.



Respuesta de fase.



No hay terminos en el denominador por lo tanto son sistemas no amortiguados

3) a)

Partiendo de la MAT hillo los nodos de Administraciones para:

Source común ($V_s = \phi$)

$$\begin{pmatrix} I_g \\ I_d \\ I_s \end{pmatrix} = \begin{pmatrix} sC_{iss} & -sC_{gd} & s(C_{gd} - C_{iss}) \\ g_m - sC_{gd} & sC_{oss} & s(C_{gd} - C_{oss}) - g_m \\ \cancel{s(C_{gd} - C_{iss}) - g_m} & \cancel{s(C_{gd} - C_{oss})} & \cancel{s(C_{iss} - C_{gd}) + s(C_{oss} - C_{gd}) + g_m} \end{pmatrix} \begin{pmatrix} V_g \\ V_D \\ \cancel{V_s} \end{pmatrix}$$

$$\begin{pmatrix} I_g \\ I_d \end{pmatrix} = \begin{pmatrix} sC_{iss} & -sC_{iss} \\ g_m - sC_{iss} & s(C_{oss}) \end{pmatrix} \begin{pmatrix} V_g \\ V_D \end{pmatrix}$$

Drain común ($V_D = \phi$)

$$\begin{pmatrix} I_g \\ I_d \\ I_s \end{pmatrix} = \begin{pmatrix} sC_{iss} & -sC_{gd} & s(C_{gd} - C_{iss}) \\ \cancel{g_m - sC_{gd}} & \cancel{sC_{oss}} & \cancel{s(C_{gd} - C_{oss}) - g_m} \\ s(C_{gd} - C_{iss}) - g_m & \cancel{s(C_{gd} - C_{oss})} & \cancel{s(C_{iss} - C_{gd}) + s(C_{oss} - C_{gd}) + g_m} \end{pmatrix} \begin{pmatrix} V_g \\ V_D \\ V_s \end{pmatrix}$$

$$\begin{pmatrix} I_g \\ I_s \end{pmatrix} = \begin{pmatrix} sC_{iss} & s(C_{oss} - C_{iss}) \\ s(C_{iss} - C_{iss}) - g_m & s(C_{iss} - C_{iss}) + s(C_{oss} - C_{iss}) + g_m \end{pmatrix} \begin{pmatrix} V_g \\ V_s \end{pmatrix}$$

b) Matriz de parámetros ABCD en source común.

$$\begin{pmatrix} I_g \\ I_d \end{pmatrix} = \begin{pmatrix} s(C_{gs} + C_{gd}) & -sC_{gd} \\ g_m - sC_{gd} & s(C_{ds} + C_{gd}) \end{pmatrix} \begin{pmatrix} V_g \\ V_d \end{pmatrix} \quad \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$$

Parámetros T

$$V_1 = AV_2 + B(-I_2)$$

$$I_1 = CV_2 + D(-I_2)$$

Parámetros Y

$$I_1 = Y_{11}V_1 + Y_{21}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

$$* A = \left. \frac{V_1}{V_2} \right|_{I_2 = \phi}$$

$$-Y_{21}V_1 = Y_{22}V_2$$

$$\frac{V_1}{V_2} = -\frac{Y_{22}}{Y_{21}}$$

$$A = -\frac{s(C_{ds} + C_{gd})}{g_m - sC_{gd}}$$

$$* B = \left. \frac{V_1}{-I_2} \right|_{V_2 = \phi}$$

$$\frac{V_1}{-I_2} = \frac{-1}{Y_{21}}$$

$$B = \frac{-1}{g_m - sC_{gd}}$$

$$* C = \left. \frac{I_1}{V_2} \right|_{I_2 = \phi}$$

$$I_1 = Y_{11}V_1 + Y_{12}V_2$$

$$0 = Y_{21}V_1 + Y_{22}V_2 \rightarrow -\frac{Y_{22}V_2}{Y_{21}} = V_1$$

$$I_1 = \left(-\frac{Y_{22}}{Y_{21}}Y_{11} + Y_{12} \right) V_2$$

$$\frac{I_1}{V_2} = Y_{12} - \frac{Y_{22}Y_{11}}{Y_{21}}$$

$$C = -sC_{gd} - \frac{s(C_{ds} + C_{gd}) + s(C_{gs} + C_{gd})}{g_m - sC_{gd}}$$

$$C = -sC_{gd} - \frac{s(C_{ds} + C_{gs} + ZC_{gd})}{g_m - sC_{gd}} = \frac{-sC_{gd}(g_m - sC_{gd}) - s(C_{ds} + C_{gs} + ZC_{gd})}{g_m - sC_{gd}}$$

$$* D = \frac{I_1}{-I_2} \Big|_{V_2=0} = \frac{\cancel{y_{11}V_1} + \cancel{y_{21}V_2}}{-(\cancel{y_{21}V_1} + \cancel{y_{22}V_2})} = -\frac{y_{11}}{y_{21}}$$

$$D = -\frac{s(C_{gs} + C_{gd})}{g_m - sC_{gd}}$$

$$T = \frac{-1}{g_m - sC_{gd}} \begin{pmatrix} s(C_{ds} + C_{gd}) & 1 \\ sC_{gd}(g_m - sC_{gd}) + s(C_{ds} + C_{gs} + ZC_{gd}) & s(C_{gs} + C_{gd}) \end{pmatrix}$$

$$T = \frac{1}{g_m - sC_{gd}} \begin{pmatrix} s(C_{oss}) & 1 \\ sC_{gd}(g_m - sC_{gd}) + s(C_{iss} + C_{oss}) & s(C_{iss}) \end{pmatrix}$$

$$T = \frac{1}{g_m - sC_{rss}} \begin{pmatrix} sC_{oss} & 1 \\ s(C_{iss} + C_{oss} + C_{rss}^2) + s g_m C_{rss} & sC_{iss} \end{pmatrix}$$